

Master Plan for Embedded Systems and Algorithm Study

Your Name

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1. Data Structures and Algorithms (Independent Study)

1.1 Algorithms

- **Fast Fourier Transform (FFT):** Study the algorithm for frequency-domain analysis, focusing on efficient computation in real-time systems.
 - **Priority:** High (for real-time signal processing with ADC)
- **Sliding Window Algorithm:** Used for maintaining a subset of items (e.g., moving average or window-based data aggregation) in real-time data streams.
 - **Priority:** High (for data stream handling and pattern detection)
- **Pattern Matching:**
 - **KMP Algorithm:** Knuth-Morris-Pratt for efficient string matching, especially useful in packet analysis or command recognition in embedded streams.
 - **Rabin-Karp Algorithm:** Used for detecting patterns in streams of data (e.g., network packet analysis).
 - **Priority:** Medium (for DMA data streaming and pattern detection)

1.2 Data Structures

- **Trie:** Efficient for storing and querying dynamic sequences or strings, useful for embedded systems with limited memory.
 - **Priority:** Medium (for managing binary streams or sequences in data buffers)
- **B-tree:** Balanced tree structure used for large datasets, suitable for file systems or large timestamp storage.

- **Priority:** Medium (for real-time clock tasks and flash memory management)
- **Binary Search Tree:** Fundamental structure for organizing and querying data efficiently.
 - **Priority:** High (for managing timestamps, task scheduling)
- **Priority Queue:** Crucial for scheduling tasks, managing sensor data, or tracking highest/lowest readings.
 - **Priority:** High (integrates with most embedded systems projects)
- **Stacks and Queues:** Basic structures useful for handling interrupts, task scheduling, and managing function calls.
 - **Priority:** Medium (for interrupt-driven task management)
- **HashMap:** Efficient for storing key-value pairs, useful in fault categorization, configuration settings, or power monitoring.
 - **Priority:** Medium (for memory management and fault handling)

2. Integrated Embedded Systems Tasks (Hardware + Algorithms)

2.1 Priority 1: Foundation-Level Tasks

- **Temperature and Voltage Monitoring:**
 - **Data Source:** Internal temperature sensor, voltage reference (VREFINT)
 - **Data Structures:** Ring Buffer, Priority Queue
 - **Algorithms:** Moving Average, Exponential Smoothing, Threshold Detection
- **Watchdog Timers:**
 - **Data Source:** Independent and Window Watchdog Timers (IWDG, WWDG)
 - **Data Structures:** Circular Queue
 - **Algorithms:** Timeout Management, Monitoring and Alerting
- **Real-Time Clock (RTC) & Timestamps:**
 - **Data Source:** RTC, SysTick timer
 - **Data Structures:** Binary Search Tree, B-tree
 - **Algorithms:** Scheduling Algorithms (Round-robin, priority-based), Binary Search

2.2 Priority 2: Intermediate-Level Tasks

- **Interrupt-Driven Task Scheduling:**
 - **Data Source:** Timers, interrupts
 - **Data Structures:** Priority Queue, Doubly Linked List
 - **Algorithms:** Round-Robin Scheduling, Earliest Deadline First (EDF)
- **Real-Time Signal Processing with ADC:**
 - **Data Source:** Analog-to-Digital Converter (ADC)
 - **Data Structures:** Circular Queue, FFT Buffer
 - **Algorithms:** FFT, Sliding Window, Finite State Machine (FSM) for pattern recognition
- **Fault Monitoring and Recovery:**
 - **Data Source:** Non-Maskable Interrupt (NMI), system faults
 - **Data Structures:** HashMap, Circular Queue
 - **Algorithms:** Error Detection (Hamming Code), CRC

2.3 Priority 3: Advanced-Level Tasks

- **DMA for Data Streaming:**
 - **Data Source:** DMA controller, ADC
 - **Data Structures:** Ring Buffer, Trie
 - **Algorithms:** Pattern Matching (KMP, Rabin-Karp), Circular Buffer Management
- **Memory Management and Flash Programming:**
 - **Data Source:** Flash memory, EEPROM
 - **Data Structures:** HashMap, Linked List
 - **Algorithms:** Wear-Leveling, CRC for Data Integrity
- **Power Consumption Monitoring:**
 - **Data Source:** Internal power sensors
 - **Data Structures:** Priority Queue, HashMap
 - **Algorithms:** Power Optimization (Dynamic Voltage/Frequency Scaling), Energy Profiling

3. Summary of Priorities

- **High Priority:**
 - **Data Structures & Algorithms:** FFT, Sliding Window, Binary Search Tree, Priority Queue.
 - **Embedded Tasks:** Temperature and voltage monitoring, watchdog timers, RTC-based scheduling, interrupt-driven task scheduling.
- **Medium Priority:**
 - **Data Structures & Algorithms:** Trie, B-tree, Pattern Matching (KMP, Rabin-Karp), HashMap.
 - **Embedded Tasks:** Fault monitoring, real-time signal processing, DMA data streaming.
- **Low Priority:**
 - Memory Management, Power Consumption Monitoring.